Curriculum Vitae

Francesco De Luca

General Information

Nazionality: Italian
Current position: PhD student of the XXXIX Cycle of the ICT doctoral program at DIMES, University of Calabria, Rende (CS), Italy
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1 Biographical notes, education, and positions held

- On Dec. 6, 2021, he received a Bachelor's degree in computer engineering with Honors from the University of Calabria, discussing a thesis entitled "Microservice Architectures and Service Mesh".
- On Dec. 16, 2022, he was awarded as *Student of Excellence* with the "Percorso di eccellenza" grant in recognition of his outstanding academic record in terms of number of exams and average grade achieved at the DIMES Department of the University of Calabria.
- On Oct. 16, 2023, he received a Master's degree in Computer Engineering, specialization in Artificial Intelligence and Machine Learning, with Honors from the University of Calabria, discussing the thesis entitled "Study and experimentation of Negative Learning approaches for Anomaly Detection".
- He is a Ph.D. student in *Information and Communication Technology* at the Department of Computer, Electronic and Systems Engineering at the University of Calabria from November 1, 2023, as a winner of the competition *D.M. 118/2023 - Pubblica Amministrazione - Digitalizzazione della Pubblica Amministrazione*. Prof. Fabrizio Angiulli and Prof. Fabio Fassetti supervise research activities.
- From March 1, 2026, to August 31, 2026, he was a visiting Ph.D. student at the University of Southern Denmark, under the supervision of Prof. Arthur Zimek. During his research stay, he worked on machine learning approaches for dialysis failure detection, counterfactual explanations, and time series analysis.

2 Education

- Master's Degree in Computer Engineering (Artificial Intelligence and Machine Learning), University of Calabria, Rende (CS), Italy.

  Thesis title: *Studying and experimenting with Negative Learning approaches for Anomaly Detection.*

  Supervisor: Prof. Fabrizio Angiulli, Co-supervisor: Dr. Luca Ferragina.

  Graduation grade: 110/110 Cum Laude.

  Date of Achievement: 16/10/2023.

– Bachelor's Degree in Computer Engineering, University of Calabria, Rende (CS), Italy.

Thesis title: *Microservice Architectures and Service Mesh.*

Supervisor: Prof. Sergio Flesca.

Graduation grade: 110/110 Cum Laude.

Date of Achievement: 06/12/2021.

3 Honors and Awards

– **Top *5* Best Papers** – 19th International Conference on Similarity Search and Application, 14-16 October, Brno, Czech Republic. **Invited to submit an extended version to a Special Issue of the *Information Systems* Journal**;

– **Best Project proposal award** – Doctoral School "Artificial Intelligence for a Secure Society", for the project proposal *Harnessing LLM Agents and RAG for Trustworthy Fake News Detection*, 9-10 September 2024, Ricadi (VV), Italy

– **Student of Excellence** – "Percorso di eccellenza" grant, for academic merits at the DIMES Department of the University of Calabria, Rende (CS), Italy.

4 Education activities

In his academic career, Francesco De Luca has acquired engineering and computer science skills. In his Master's degree course, he specialized in Artificial Intelligence and Machine Learning, with particular interest in sub-symbolic techniques. During the same course, he has developed many teaching projects in different areas of Artificial Intelligence, including *Computer Vision* (Diffusion Models for Conditional Face Generation), *Social Network and Media Analysis* (A Multimodal Approach for MEME Classification) and *Deep Learning* in general (Multi-task Learning), and acquired skills including data analysis and the use of frameworks such as TensorFlow and PyTorch. During his Ph.D., he participated in numerous teaching activities, including doctoral courses, seminars, and workshops; he also participated in a doctoral school and was the winner of the related competition: *Best Project proposal - Artificial Intelligence for a Secure Society.*

5 Research activities

The main research area of Francesco De Luca concerns machine learning systems and, in particular, methodologies related to neural networks. In his Master's thesis in Computer Engineering, he worked on unsupervised techniques for Anomaly Detection. During his Ph.D., he has focused on *Explainable Artificial Intelligence*, *Anomaly Detection*, and *Continual Learning.* The following is a summary description of the directions followed for each line of research.

- *Explainable Artificial Intelligence*: he has investigated techniques of Explainable Artificial Intelligence for textual classifiers. Specifically, he proposed a *post-hoc* approach based on *Interpretable Local Surrogates*, introducing a technique to generate neighborhoods for training local surrogates and to create more interpretable explanations and counterfactuals [3], [1]. He has also proposed an *attention-mechanism*-based method for computing token aggregations from attention maps to build explanations for *transformer*-based classifiers [4].

- *Anomaly Detection in Time Series*: he has investigated using *Large Language Models* for anomaly detection in univariate time series, exploring LLMs' numerical sequence processing and generalization capabilities. The research was conducted using a zero-shot and unsupervised approach, with the result confirming the reliability of pretrained language models for numerical sequence analysis [5].

- *Anomaly Detection in Continual Learning*: he has investigated the intersection between *Continual Learning* and *Anomaly Detection*. He carried out a comprehensive survey on *Continual Anomaly Detection*, providing a unified formalization of the problem, a common evaluation framework, and a taxonomy of existing methods along three dimensions, together with an analysis of application domains, benchmarks, and open challenges [2]. He has also proposed a task-agnostic framework that detects anomalies in the data stream feeding a continual learner: by maintaining an adaptive set of concepts, it distinguishes new tasks and concept drift from genuine anomalies, so that only reliable data are used to update the model, consistently improving the accuracy of state-of-the-art continual learning methods [6].

6 Professional Activities

Within his research group, Francesco De Luca is responsible for the implementation of the code of the scientific works and for the execution of the experimental campaigns. He also manages part of the research group's GitHub repository, `https://github.com/AIDALab-DIMES`, which makes publicly available the code of the group's scientific works. Specifically, he developed and maintains the following repositories:

- **LLiMe** (`https://github.com/AIDALab-DIMES/LLiMe`): official implementation of LLiMe, a Large Language Model-based local surrogate explainer for black-box textual classifiers [3], [1]. The code implements the overall framework, including the LLM querying pipeline via API, the local surrogate training, and the explanation generation.
- **TiLLMeOut** (`https://github.com/AIDALab-DIMES/TiLLMeOut`): official implementation of TiLLMeOut (Time Series Large Language Models-based Outlier Detection), a zero-shot and prompt-free anomaly detector for univariate time series, which encodes the series as text, reconstructs it through the language-modeling head of a pre-trained BART model, and scores each point by its reconstruction error, with an iterative refinement that uncovers masked anomalies[5]. The code implements the time series preprocessing, the quantization step, and the iterative reconstruction procedure.
- **ADoCL** (`https://github.com/AIDALab-DIMES/ADoCL`): official implementation of ADoCL (Anomaly-aware Dynamic Concept maintenance for reliable Continual Learning), a task-agnostic filter placed upstream of any continual learner, which maintains an adaptive index of concepts, modeled as Conditional Auto-Encoders, to separate new tasks and concept drift from genuine anomalies, forwarding only reliable batches to the learner [6]. The repository contains the C-AE-based *concept*, the *memory-buffer*, and the *ADoCL detector* implementations.
- **ECAM** (`https://github.com/AIDALab-DIMES/ECAM`): official implementation of ECAM (Ensemble Clustering from Attention Maps), which explains Transformer-based text classifiers by clustering the tokens on each attention head, combining the per-head groupings through ensemble clustering, and selecting the token clusters that best trade off comprehensiveness and sufficiency [4]. The repository implements the extraction of the attention maps, the aggregation of tokens using HDBSCAN, and the ensemble aggregation to be returned as an explanation.

All the implementations are written in Python, and they use several libraries such as *numpy*, *pandas*, *scikit-learn*, and *pytorch*.

Additionally, since 2024, Francesco has been responsible for the laboratory activities of the *Machine and Deep Learning* Master's degree course in Computer Engineering. In this role, he has developed the practical implementations and demonstration notebooks of the machine learning techniques presented in the theoretical part of the course, including linear models, support vector machines, advanced neural network architectures (CNNs, RNNs, GNNs, AEs, VAEs, and GANs), reinforcement learning, and explainable artificial intelligence.

As for technical competences, Francesco has acquired a wide range of programming and software development skills throughout his academic journey, spanning his Bachelor's degree, Master's degree, and Ph.D. studies. Over the years, he has built solid expertise in Python, mastering core data science,

deep learning, and scientific computing libraries such as *PyTorch*, *TensorFlow*, *scikit-learn*, *pandas*, and *NumPy*. Beyond Python, his background encompasses strong competencies in compiled and general-purpose programming languages including *Java* and *C*, as well as shell scripting with *Bash* for environment configuration, task automation, and experimental workflow management in Linux environments.

7 List of Publications

In all publications listed, all authors contributed equally to all stages of writing.

Journals

[1] F. Angiulli, F. De Luca, F. Fassetti, and S. Nisticò, "Llime: Enhancing text classifier explanations with large language models," *Machine Learning*, vol. 114, no. 12, p. 271, 2025.

[2] F. Angiulli, F. De Luca, and F. Fassetti, "A comprehensive survey on continual anomaly detection," *Neurocomputing*, vol. 707, p. 135 118, 2027, ISSN: 0925-2312. DOI: `https://doi.org/10.1016/j.neucom.2026.135118`. [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S0925231226025166`.

Conferences

[3] F. Angiulli, F. De Luca, F. Fassetti, and S. Nisticò, "Large language models-based local explanations of text classifiers," in *Discovery Science*, D. Pedreschi, A. Monreale, R. Guidotti, R. Pellungrini, and F. Naretto, Eds., Cham: Springer Nature Switzerland, 2025, pp. 19–35, ISBN: 978-3-031-78977-9.

[4] F. Angiulli, F. De Luca, and F. Fassetti, "Attention to those tokens! ensemble clustering can explain attention-based text classifiers.," in *CIKM*, ACM, 2026.

[5] F. Angiulli, F. De Luca, and F. Fassetti, "Tillmeout: Time series large language models-based outlier detection," in *Discovery Science*, M. Cerrato, J. N. van Rijn, and S. Kramer, Eds., Cham: Springer Nature Switzerland, 2027, pp. 249–264, ISBN: 978-3-032-41086-3.

[6] F. Angiulli, F. De Luca, and F. Fassetti, "Adocl: Anomaly-aware dynamic concept maintenance for reliable continual learning," in *Similarity Search and Applications*, K. Schoeffmann, F. Carrara, V. Mic, *et al.*, Eds., Cham: Springer Nature Switzerland, 2027, pp. 198–213, ISBN: 978-3-032-42224-8.

8 Participation as a Speaker at Conferences of International Interest

Francesco De Luca has been a speaker at the following international conferences and has presented the indicated papers:

- *19th International Conference on Similarity Search and Applications* (SISAP 2026), Brno, Czech Republic, October 14-16 2026, presenting the work *ADoCL: Anomaly-aware dynamic concept maintenance for reliable continual learning* [6].
- *29th International Conference on Discovery Science* (DS 2026), Mainz, Germany, October 5-9 2026, presenting the work *TiLLMeOut: Time series large language models-based outlier detection* [5].
- *27th International Conference on Discovery Science* (DS 2024), Pisa, Italy, October 14-16 2024, presenting the work *Large Language Models-based Local Explanation of Text Classifier* [3].

9 Activities as Reviewer

Participation in program committees

Francesco De Luca is a member of the program committee of:

- AIxIA Workshop on Green-Aware Artificial Intelligence (Green-Aware AI 2024), November 25-28 2024, Bolzano (BZ), Italy
- ADIBIS Workshop on Cooperative AI Models and Applications (CAIMA 2025), September 23 2025, Tampere, Finland.

Reviews for journals

2026 *International Journal of Nephrology*, Wiley (Hindawi).

2024–2026 *AI Communications: The European Journal on Artificial Intelligence*, IOS Press.

2024–2026 *Scientific Reports*, Springer Nature.

Reviews for conferences

2026 ECML/PKDD 2026: European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, September 07 - September 11 2026, Naples

2026 ACM-BCB 2026: 17th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics June 30 - July 3 2026, Rende

2025 ECML/PKDD 2025: European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, September 15 - September 19 2026, Porto

2025 IJCNN 2025: International Joint Conference on Neural Networks, June 30 - July 5 2025, Rome

2025 ICHI 2025: IEEE-International Conference on Healthcare Informatics, June 18-21, 2025, Rende

2024 ICDM 2024: IEEE International Conference on Data Mining, December 9-12 2024, Abu Dhabi

2024 ECML/PKDD 2024: European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, September 9-13 2024, Vilnius

2024 PAKDD 2024: Pacific-Asia Conference on Knowledge Discovery and Data Mining, May 7-10 2024, Taipei

10 Teaching activities

Francesco De Luca has teaching experience in both computer engineering and artificial intelligence. His teaching activities have covered topics in Machine and Deep Learning, including linear models, Support Vector Machines, advanced neural network architectures (CNNs, RNNs, GNNs, AEs, VAEs, and GANs), Reinforcement Learning, and Explainable Artificial Intelligence. He has also taught Logical Networks and Computers, covering finite automata, Karnaugh maps, accumulator machines, RTL, and Assembly language programming.
Specifically, he has been a teaching assistant in the following courses:

Academic Year 2026/2027

- *Deep Learning*, 16 hours, Master's degree in Computer Engineering, DIMES, University of Calabria, Rende (CS).

Academic Year 2025/2026

– *Machine and Deep Learning*, 16 hours, Master's degree in Computer Engineering, DIMES, University of Calabria, Rende (CS).

Academic Year 2024/2025

– *Machine and Deep Learning*, 16 hours, Master's degree in Computer Engineering, DIMES, University of Calabria, Rende (CS).

Academic Year 2023/2024

– *Logical Networks and Computers*, 27 hours, Bachelor's Degree in Computer Engineering, DIMES, University of Calabria, Rende (CS).

11 Thesis supervision

Bachelor's Degree

2024 *Large Language Models-based Image Reconstruction for Anomaly Detection.* Bachelor's Degree in Computer Engineering, DIMES, University of Calabria. Candidate: Anna Chiara Bruni. Supervisor: Prof. Fabio Fassetti, Co-supervisor: Dr. Francesco De Luca.

Master's Degree

2025 *Deep Learning Techniques for analysis and classification of electrocardiograms.* Master's Degree in Artificial Intelligence and Computer Science, DEMACS, University of Calabria. Candidate: Giovambattista Crudo. Supervisors: Prof. Valeria Fionda, Prof. Fabio Fassetti, Co-supervisor: Dr. Francesco De Luca.

2025 *Outlier detection on time series using LLM with long-context input.* Master's Degree in Computer Engineering, DIMES, University of Calabria. Candidate: Simone Caravetta. Supervisors: Prof. Fabrizio Angiulli, Co-supervisor: Dr. Francesco De Luca.

I, the undersigned, Francesco De Luca, hereby authorise the processing of my personal data contained in this curriculum vitae pursuant to Legislative Decree 196/2003 and Regulation (EU) 2016/679 (GDPR).

I, the undersigned, Francesco De Luca, hereby declare that the information contained in this curriculum vitae is true and accurate, pursuant to Article 76 of Presidential Decree (D.P.R.) 445/2000.